Enable Logs for a Component

Introduction

Logging is one of the most important aspects of a production-grade server. A properly configured logging system is vital for identifying errors, security threats, and usage patterns.

Out of the log files which are created for each log type, the wso2carbon log covers all the management features of a product.

In this exercise, we're enabling DEBUG logs for a given component. In the existing `log4j2.properties` file, some loggers are already configured, and changing the log level to the desired value is sufficient. For the ones that are not configured, logger configuration should be added first.

WSO2 Server restart is not required for these changes.

Configurations

See the following instructions on how to enable debug logs for a component in WSO2 Identity Server:

Enable DEBUG logs for org.wso2.carbon.user.core

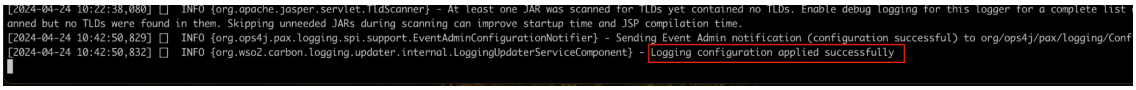
Follow the steps given below.

1. Navigate to the `<IS_HOME>/repository/conf/log4j2.properties` file.
2. Change the log level from logger.org-wso2-carbon-user-core.level=**INFO** to logger.org-wso2-carbon-user-core.level=**DEBUG.**
3. Then you need to append the org-wso2-carbon-user-core logger to the logger list.

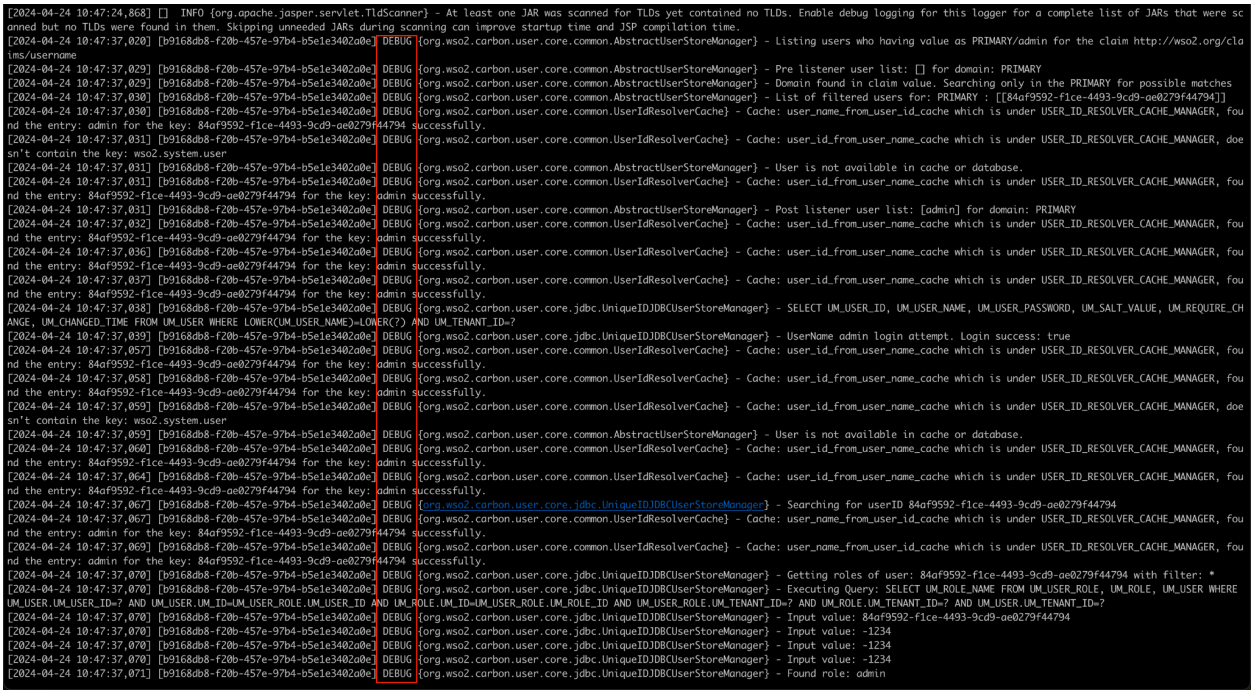
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- For this package, loggers are already configured. This can be confirmed by searching for org-wso2-carbon-user-core in the *log4j2.properties* file.
- You should see 3 occurrences.
- Therefore, changing the log level is sufficient.

4. After saving the `log4j2.properties` file, see that the following log lines are printed to the wso2carbon logs confirming that the configuration change was applied successfully.



After this, WSO2 IS prints DEBUG logs for the org.wso2.carbon.user.core package in the wso2carbon logs whenever a user-related operation runs, for example when you log in to the Console or list users. The screenshot below shows an example.



Enable DEBUG logs for org.wso2.carbon.identity.handler.event.account.lock

Follow the steps given below.

1. Navigate to the `<IS_HOME>/repository/conf/log4j2.properties` file

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2. Search for the `org.wso2.carbon.identity.handler.event.account.lock` package and see that it is not there. This package locks and unlocks user accounts.
3. First, configure the logger in the following format:

```
logger.<LOGGER_NAME>.name=<PACKAGE_NAME>
logger.<LOGGER_NAME>.level=<LOG_LEVEL>
```

   Here, the LOGGER_NAME has to be a unique string and does not need to be the same as the package name.

```
logger.account-lock-
handler.name=org.wso2.carbon.identity.handler.event.account.lock
logger.account-lock-handler.level=DEBUG

logger.email-
adapter.name=org.wso2.carbon.event.output.adapter.email
logger.email-adapter.level=DEBUG
```

4. The second logger (`email-adapter`) is optional: it shows the notification email being sent. After this step, add the logger names to the beginning of the existing `loggers` line, separated by commas. Do not remove the names that are already there.

```
loggers = account-lock-handler, email-adapter, AUDIT_LOG, trace-
messages, diagnostics, correlation, … (keep the rest of the
existing list)
```

5. Save the file.

Wait until `Logging configuration applied successfully` is printed in the logs (no restart is needed). Then, in the Console, go to **User Management > Users**, select a user that has an email address, and turn on **Lock user** (confirm). Turn it off again to unlock the user. Each time, WSO2 IS prints DEBUG lines like the ones below: `User <username> is locked` / `User <username> is unlocked` from the account lock handler, and the lock or unlock notification email being sent.

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```
[2026-11-08 21:10:38,278] DEBUG {org.wso2.carbon.identity.handler.event.account.lock.AccountLockHandler} - User tommy is locked
[2026-11-08 21:10:38,336] DEBUG {org.wso2.carbon.event.output.adapter.email.EmailEventAdapter} - Attempting to send an email to tommy@wso2.com
[2026-11-08 21:10:38,454] DEBUG {org.wso2.carbon.event.output.adapter.email.EmailEventAdapter} - Mail sent to the EmailID tommy@wso2.com Successfully
[2026-11-08 21:10:52,846] DEBUG {org.wso2.carbon.identity.handler.event.account.lock.AccountLockHandler} - User tommy is unlocked
[2026-11-08 21:10:52,852] DEBUG {org.wso2.carbon.event.output.adapter.email.EmailEventAdapter} - Attempting to send an email to tommy@wso2.com
[2026-11-08 21:10:52,920] DEBUG {org.wso2.carbon.event.output.adapter.email.EmailEventAdapter} - Mail sent to the EmailID tommy@wso2.com Successfully
```

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